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Insurance Charges Machine Learning

Objective

The objective of this project was to apply core machine learning concepts to a real-world regression problem. Specifically, the goal was to build and compare multiple models to predict medical insurance charges while gaining practical experience with data preprocessing, feature engineering, model evaluation, and interpretation. Through this process, the project emphasizes understanding how different modeling approaches affect predictive performance and insight generation.

The Dataset

The dataset used in this project contains medical insurance cost information for 1,338 individuals in the United States. It includes demographic and health-related variables such as age, sex, Body Mass Index (BMI), number of children, smoking status, and residential region. The target variable, charges, represents the medical insurance cost billed to each individual. This dataset is commonly used for regression modeling and provides a structured framework for analyzing how lifestyle and demographic factors influence healthcare expenses.

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520
...	...	...	...	...	...	...	...
1333	50	male	30.970	3	no	northwest	10600.54830
1334	18	female	31.920	0	no	northeast	2205.98080
1335	18	female	36.850	0	no	southeast	1629.83350
1336	21	female	25.800	0	no	southwest	2007.94500
1337	61	female	29.070	0	yes	northwest	29141.36030

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   age               1338 non-null   int64
 1   sex               1338 non-null   int64
 2   bmi               1338 non-null   int64
 3   children          1338 non-null   int64
 4   smoker            1338 non-null   int64
 5   charges           1338 non-null   int64
 6   region_northwest  1338 non-null   int64
 7   region_southeast  1338 non-null   int64
 8   region_southwest  1338 non-null   int64
dtypes: int64(9)
memory usage: 94.2 KB
```

	age	sex	bmi	children	smoker	charges	region_northwest	region_southeast	region_southwest
count	1338.000000	1338.000000	1338.000000	1338.000000	1338.000000	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	0.505232	30.169656	1.094918	0.204783	13269.928999	0.242900	0.272048	0.242900
std	14.049960	0.500160	6.119875	1.205493	0.403694	12110.012755	0.428995	0.445181	0.428995
min	18.000000	0.000000	15.000000	0.000000	0.000000	1121.000000	0.000000	0.000000	0.000000
25%	27.000000	0.000000	26.000000	0.000000	0.000000	4740.000000	0.000000	0.000000	0.000000
50%	39.000000	1.000000	30.000000	1.000000	0.000000	9381.500000	0.000000	0.000000	0.000000
75%	51.000000	1.000000	34.000000	2.000000	0.000000	16639.250000	0.000000	1.000000	0.000000
max	64.000000	1.000000	53.000000	5.000000	1.000000	63770.000000	1.000000	1.000000	1.000000

Goals

01

Develop a Strong Understanding of Regression Modeling

Apply and compare multiple regression techniques, including linear regression, polynomial regression, and Gradient Boosting, to understand how different models capture relationships within structured data.

02

Practice End-to-End Machine Learning Workflow

Gain hands-on experience with data preprocessing, exploratory data analysis, feature engineering, model training, evaluation, and interpretation in a complete machine learning pipeline.

03

Interpret Model Results and Extract Insights

Analyze feature importance and residual patterns to identify the key drivers of insurance costs and understand how demographic and health-related factors influence predictions.

Workflow

- Data Cleaning / Preprocessing
- Exploratory Data Analysis / Visualization
- Linear Regression
- Polynomial Regression
- Gradient Boosting Regressor
- Model Comparison & Interpretation

Data Cleaning / Preprocessing

```python
# Let's change the sex, smoker, and region columns into numerical data
```

```python
# Sex
df["sex"] = df["sex"].map({"female": 0, "male": 1})
```

```python
# 0 = female, 1 = male
df["sex"].value_counts()
```

```
sex
1    676
0    662
Name: count, dtype: int64
```

```python
# smoker
df["smoker"] = df["smoker"].map({"no": 0, "yes": 1})
```

```python
# 0 = not a smoker, 1 = smoker
df["smoker"].value_counts()
```

```
smoker
0    1064
1     274
Name: count, dtype: int64
```

```python
# region - One Hot Encoding
df["region"].value_counts()
```

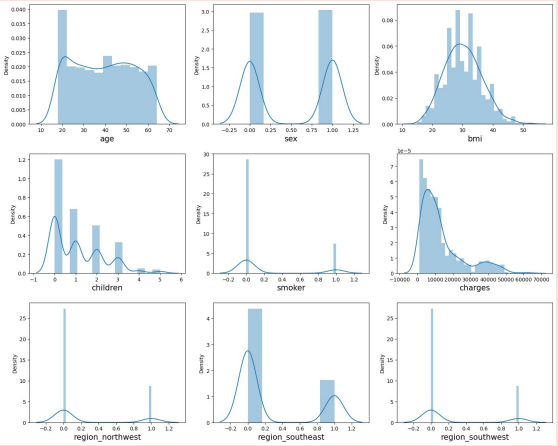
```
region
southeast    364
southwest    325
northwest    325
northeast    324
Name: count, dtype: int64
```

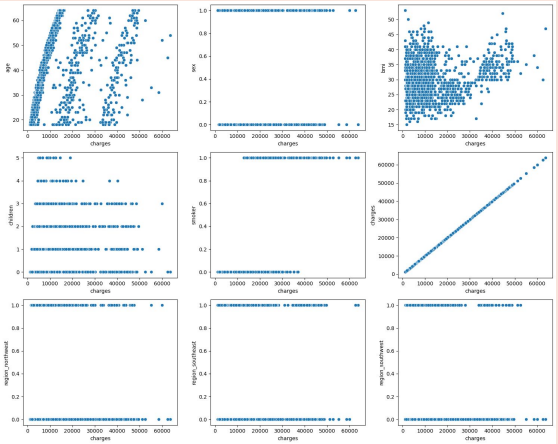
```python
df = pd.get_dummies(df, columns=["region"], drop_first=True).astype(int)
```

- Converted categorical features (sex, smoker) to numeric using mapping.
- One-hot encoded the region variable with drop_first=True.
- Checked value counts to ensure encoding correctness.
- Standardized features using StandardScaler (scaling before train/test split is slightly suboptimal—fit should occur on training data only).

Exploratory Data Analysis / Visualization

- Plotted distributions of all features to assess skewness, outliers, and data ranges.
- Scatter plots of each feature vs. target (charges) to identify potential linear/nonlinear relationships.
- Observations highlighted how age, BMI, and smoking status correlate with insurance costs.
- Created additional plots showing feature interactions, e.g., BMI vs. charges by smoker status.





Linear Regression

- Split dataset into train/test sets.
- Fit a baseline Linear Regression model.
- Calculated R^2 and adjusted R^2 for train and test sets.
- Checked multicollinearity via VIF.
- F-test analysis to identify statistically significant predictors.
- Coefficient analysis to assess feature contributions.
- Residual analysis to check linear model assumptions and detect nonlinearity/heteroscedasticity.

Calculated Variance Inflation Factor (VIF) scores to check for multicollinearity among predictors. VIFs close to 1 indicate that the features are not highly correlated, so we safely use them in regression without worrying about instability in the estimated coefficients.

	VIF	Features
0	1.016636	age
1	1.008930	sex
2	1.107868	bmi
3	1.003971	children
4	1.012149	smoker
5	1.518821	region_northwest
6	1.653430	region_southeast
7	1.529859	region_southwest

```python
# Regression
X = df.drop(columns = "charges", axis = 1)
y = df["charges"]
```

```python
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_scaled
```

Linear Regression is a supervised learning algorithm used for predicting continuous outcomes. It models the relationship between independent variables and a target variable by fitting a linear equation to the data. The model estimates coefficients that quantify how much the target variable changes with a one-unit change in each predictor, assuming all other variables remain constant.

```python
# splitting data into training and test set
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size = 0.3, random_state = 0)
```

```python
# fitting the training data into model
from sklearn.linear_model import LinearRegression
lr = LinearRegression()
lr.fit(X_train, y_train)
```

```
LinearRegression
LinearRegression()
```

```python
# prediction of model
y_predicted = lr.predict(X_test)
```

```python
# prediction of model
y_predicted = lr.predict(X_test)
```

```python
# training accuracy of model -R^2
lr.score(X_train, y_train)
```

```python
# test accuracy of model -R^2
lr.score(X_test, y_test)
```

```python
# creating a function to create adjusted R-Squared

def adj_r2(X, y, model):
    r2 = model.score(X, y)
    n = X.shape[0]
    p = X.shape[1]
    adjusted_r2 = 1 - (1 - r2) * (n - 1) / (n - p - 1)

    return adjusted_r2
```

```python
print(adj_r2(X_train, y_train, lr))
```

```python
print(adj_r2(X_test, y_test, lr))
```

The full F-test and coefficient analysis show that smoking status, age, and BMI are the most statistically significant and influential predictors of insurance charges. Smoking has the largest positive effect, substantially increasing predicted costs, while age and BMI also contribute strongly to higher charges. Regional variables and number of children show smaller effects, and sex has minimal impact. Overall, demographic and health-related factors, particularly smoking, drive insurance costs far more than geographic differences.

```python
X_features = X_train[:, feature_indices]

f_stats, p_values = f_regression(X_features, y_train)
feature_importance = pd.DataFrame({
    "Feature": features,
    "P-value": p_values
}).sort_values("P-value")
print(feature_importance)
```

```
             Feature        P-value
4             smoker  9.831220e-185
0                age   3.025059e-18
2                bmi   4.346584e-08
6  region_southeast   6.010482e-03
3           children   1.836582e-02
7  region_southwest   2.764666e-02
1                sex   1.280704e-01
5  region_northwest   2.266294e-01
```

The F-test results indicate that smoking status, age, and BMI are the most statistically significant predictors of insurance charges, showing the strongest evidence of meaningful relationships with cost. Region_southeast, children, and region_southwest are also statistically significant at conventional levels, but their influence is notably weaker. Sex and region_northwest are not statistically significant, suggesting they do not meaningfully improve the model's explanatory power. Overall, health-related and demographic factors contribute far more to variation in charges than regional or sex variables.

The regression coefficients show that smoking status has the largest positive effect on insurance charges, increasing predicted costs substantially when all other variables are held constant. Age and BMI also have strong positive relationships with charges, indicating that higher values in these variables correspond to higher predicted costs. The number of children has a modest positive effect. Living in the southeast, southwest, or northwest slightly decreases predicted charges relative to the baseline region, while sex has a negligible impact. Collectively, smoking, age, and BMI are the primary drivers of insurance costs in this model.

```python
# Let's calculate feature importance
features = ["age", "sex", "bmi", "children", "smoker",
            "region_northwest", "region_southeast",
            "region_southwest"]

coefs = lr.coef_

importance_df = pd.DataFrame({
    "Feature": features,
    "Coefficient": coefs,
    "AbsValue": abs(coefs)
}).sort_values("AbsValue", ascending=False)
print(importance_df[["Feature", "Coefficient"]])
```

```
             Feature  Coefficient
4             smoker  9463.836661
0                age  3603.267140
2                bmi  2050.053670
3           children   575.677431
6  region_southeast  -450.721653
7  region_southwest  -347.095170
5  region_northwest  -239.884581
1                sex   -22.709637
```

Polynomial Regression

```python
# Expanding features to include all pairwise interactions so the model can capture combined effects
X = df.drop(columns="charges")
y = df["charges"]
```

```python
# Scale features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```python
# Using polynomial features to model potential non linear relationships
from sklearn.preprocessing import PolynomialFeatures

poly = PolynomialFeatures(degree=2, interaction_only=True, include_bias=False)
X_poly = poly.fit_transform(X_scaled)
```

```python
X_train, X_test, y_train, y_test = train_test_split(X_poly, y, test_size=0.3, random_state=0)
```

```python
lr_poly = LinearRegression()
lr_poly.fit(X_train, y_train)
```

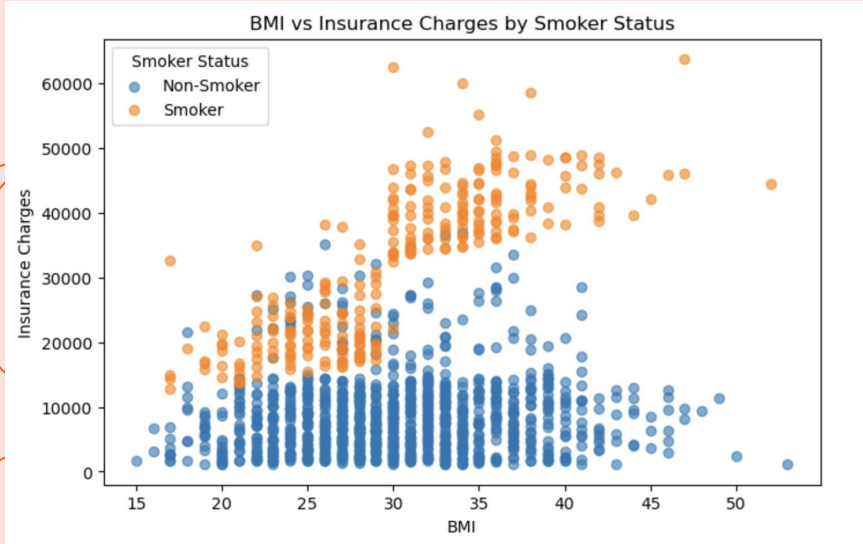
```
▾ LinearRegression
LinearRegression()
```

Polynomial Regression is an extension of linear regression that models nonlinear relationships by including polynomial terms of the predictors. It captures curvature and interactions in the data, allowing the model to better fit complex patterns while still estimating coefficients for interpretation.

- Generated polynomial interaction features (degree=2, interaction_only=True) to capture feature interactions like BMI × smoker.
- Trained a linear model on these expanded features.
- Calculated R^2, adjusted R^2, and visualized residuals.
- Feature importance interpretation focused on combined effects.
- Demonstrated improvement over baseline linear regression in capturing nonlinear patterns.

The scatter plot of BMI versus insurance charges, separated by smoker status, shows that smokers consistently have higher costs than non-smokers, with differences growing at higher BMI values. This illustrates the combined effect of smoking and BMI on insurance charges, supporting the statistical findings.

After including all pairwise interactions, the polynomial model confirmed that smoking, age, and BMI are the main drivers of insurance charges. Interactions, such as BMI with smoking and regional effects, reveal subtle patterns and refine predictions for specific subgroups, improving interpretability beyond the main predictors.




```python
feature_names = poly.get_feature_names_out(X.columns)
```

```python
coeffs = pd.DataFrame({
    "Feature": feature_names,
    "Coefficient": lr_poly.coef_
}).sort_values("Coefficient", key=abs, ascending=False)

# Show top 15 by magnitude
print(coeffs.head(15))
```

```
                          Feature  Coefficient
4                          smoker  9659.104844
0                             age  3824.557380
22                      bmi smoker  3632.560260
2                             bmi  2036.759867
3                        children   668.438829
24            bmi region_southeast  -562.496444
13            age region_southeast   380.540408
25            bmi region_southwest  -378.384908
14            age region_southwest   354.364427
29       children region_southwest  -323.678925
35  region_southeast region_southwest   312.964421
18            sex region_northwest   306.085467
19            sex region_southeast   305.160020
12            age region_northwest   288.498872
7                 region_southwest  -280.431644
```

Gradient Boosting Regressor

```python
feature_names = poly.get_feature_names_out(X.columns)
```

```python
feature_importance = pd.DataFrame({
    "Feature": feature_names,
    "Importance": gbr.feature_importances_
}).sort_values("Importance", ascending=False)

print(feature_importance.head(15))
```

```
                  Feature  Importance
4                  smoker    0.657788
22             bmi smoker    0.088230
2                     bmi    0.080197
0                     age    0.063774
11            age smoker    0.050278
10           age children    0.009633
9                  age bmi    0.009539
25  bmi region_southwest    0.005181
13  age region_southeast    0.004951
23  bmi region_northwest    0.004308
15                sex bmi    0.003819
12  age region_northwest    0.003628
21           bmi children    0.003578
24  bmi region_southeast    0.002699
16           sex children    0.001790
```

```python
# Gradient Boosting can capture nonlinear relationships and more complex
from sklearn.ensemble import GradientBoostingRegressor
gbr = GradientBoostingRegressor(
    n_estimators=1000,
    learning_rate=0.01,
    min_samples_leaf=5,
    max_depth=3,
    random_state=0)
```

```python
# Fit the model on the training data
gbr.fit(X_train, y_train)
```

```
GradientBoostingRegressor
GradientBoostingRegressor(learning_rate=0.01, min_samples_leaf=5,
                          n_estimators=1000, random_state=0)
```

Gradient Boosting Regressor is an ensemble learning method that builds a sequence of decision trees to predict a continuous outcome. Each new tree is trained to correct the errors made by the previous trees by focusing on the residuals. By iteratively improving the model and combining many small trees, Gradient Boosting can capture complex nonlinear relationships and interactions between features, often leading to strong predictive performance.

- Fit Gradient Boosting to capture nonlinear relationships and complex feature interactions.
- Tuned hyperparameters to balance bias and variance (e.g., 1000 estimators, learning_rate=0.01).
- Evaluated performance via R^2, adjusted R^2, residual plots.
- Extracted feature importances to identify dominant predictors.
- Showed that Gradient Boosting captured nonlinear interactions better than polynomial regression, especially for high-risk groups (smokers with high BMI).

Gradient Boosting Regressor

```
feature_names = poly.get_feature_names_out(X.columns)
```

```
feature_importance = pd.DataFrame({
    "Feature": feature_names,
    "Importance": gbr.feature_importances_
}).sort_values("Importance", ascending=False)

print(feature_importance.head(15))
```

```
                 Feature  Importance
4                 smoker    0.657788
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2                    bmi    0.080197
0                    age    0.063774
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    random_state=0)
```

```
# Fit the model on the training data
gbr.fit(X_train, y_train)
```

```
GradientBoostingRegressor
GradientBoostingRegressor(learning_rate=0.01, min_samples_leaf=5,
                          n_estimators=1000, random_state=0)
```

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Model Comparison & Interpretation

- Summarized train/test R^2 and adjusted R^2 across all three models.
- Identified smoking status, BMI, and age as primary drivers of insurance costs.
- Showed how adding interactions and using more flexible models improved predictive accuracy and interpretability.
- Highlighted practical insights for insurance cost prediction and potential future directions.

	Model	Train R^2	Test R^2	Train Adj R^2	Test Adj R^2
0	Linear Regression	0.731	0.791	0.729	0.786
1	Polynomial Regression	0.830	0.869	0.824	0.856
2	Gradient Boosting	0.900	0.885	0.896	0.874

Model performance improves as complexity increases. The linear regression model provides a baseline, while polynomial regression improves predictions by capturing interactions such as BMI with smoking and age with smoking. Gradient Boosting achieves the strongest and most balanced performance, indicating better generalization. Feature importance across models consistently shows that smoking status is the dominant driver of insurance costs, with BMI and age also playing major roles, while children and regional factors have smaller effects.

Thank you!